

PUSHKAR KODIGEHALLI UMESH

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SUMMARY

Postgraduate in Electronic Communication and Computer Engineering (MSc, University of Nottingham; dissertation submitted, result pending) with hands-on FPGA/VHDL, embedded systems and microcontroller experience. Designed and verified a synthesisable matrix processing core on an AMD (Xilinx) FPGA; applied statistical analysis and machine learning to real GNSS and sensor data. Completed GeoAI and systems operations internships in 2026.

SKILLS

- **Hardware & Embedded:** VHDL, Verilog, FPGA, C, C++, Embedded Systems, Microcontrollers
- **Verification & Tools:** VHDL Testbench Verification, MATLAB, DSP, Communication Systems, Vivado, Git, GitHub
- **GNSS & Data:** GNSS (NMEA, RINEX), Python, Matplotlib, QGIS, Statistical Analysis, Machine Learning, MySQL
- **Other:** IoT, RFID, Arduino, Linux (basic), CCNA (fundamentals), TCP/IP

PROJECTS

Matrix Processing Core in VHDL, *University of Nottingham – HDL Coursework* 2026

- Designed a synthesisable 8×16 , 16×16 and 16×8 matrix processing core in VHDL on an AMD (Xilinx) FPGA, performing multi-stage matrix multiplication with dual multiply–accumulate (MAC) units at 5 MHz.
- Buffered input, intermediate and output data in dual-port block RAM; each multiplication took 3,970 clock cycles.
- Verified the core with an automated file-I/O VHDL testbench, checking outputs against a MATLAB reference model.

Explainable AI for GNSS Data Quality and Environment Assessment from NMEA Logs

MSc Dissertation (submitted), University of Nottingham, supervised by Dr Minglei You 2026

- Framed GNSS data-quality and environment assessment on NMEA logs as regression and classification tasks, applying explainable AI.
- Benchmarked regression (Ridge, SVM, Random Forest, Gradient Boosting) against HDOP-only and training-mean baselines, and classification (Random Forest, Gradient Boosting, SVM, Logistic Regression) against HDOP-only and majority-class baselines.

RFID Gate System & Mini Elevator, *CHRIST University – Final-year project* 2024

- Built an Arduino/RFID gate that authenticated tags and drove the gate motors, programmed for real-time I/O.
- Developed a mini elevator with button, switch and LED floor control; tested and debugged both prototypes.

WORK EXPERIENCE

GeoAI Intern, *Geospatial Ventures Limited (GVL), Nottingham* Jun – Aug 2026

- Built Python pipelines parsing RINEX and NMEA GNSS data for a soil moisture and battery monitoring deployment at Snake Pass.
- Diagnosed a base station failure via backtest analysis; ran statistical analysis and outlier removal on 22 days of fix-quality data, visualised in QGIS and Matplotlib.
- Developed ML models for soil moisture and station uptime prediction, and automated Sentinel-2 download and spectral index processing.

Systems Operationalise & SEO Campaign Manager (Internship), *ICN Media, Nottingham* Aug – Sep 2026

- Authored Standard Operating Procedures, optimised workflows and audited website SEO performance.

Data Engineer (Micro Placement), *University of Nottingham x Practera* Jun 2026

- Audited and reconciled two spreadsheet-based inventory systems, profiling 300+ records across 40+ fields to guide consolidation into a simplified data model.

EDUCATION

MSc Electronic Communication and Computer Engineering, *University of Nottingham 2025 – 2026 (result pending)*

Modules: HDL for Programmable Devices (**83%**); Advanced Computational Engineering (**85%**); Artificial Intelligence and Intelligent Systems (**77%**); Scalable Cross-Platform Software Design (**76%**).

BSc Physics, Mathematics and Electronics, *CHRIST (Deemed to be University), India* 2021 – 2024

CGPA 8.49/10 (First Class). Grades of O or A+ in Verilog and FPGA-Based Design; Microprocessor and Microcontroller; Digital Signals and System Architecture; Embedded Systems and IoT Fundamentals (C programming).